

College of Industrial Technology
King Mongkut's University of Technology North Bangkok

Seat No.

Final Examination of Semester 2

Year: 2018

Subject: 340152 Electrical Measuring Instruments

Section: 5-6

Date: 25 April 2019

Time: 10.00-12.00

Name _____ ID: _____ Class _____

Directions: The test is designed to measure your comprehension. The test is divided into 19 questions. There will be 11 pages (including this page) and they are worth 70 points.

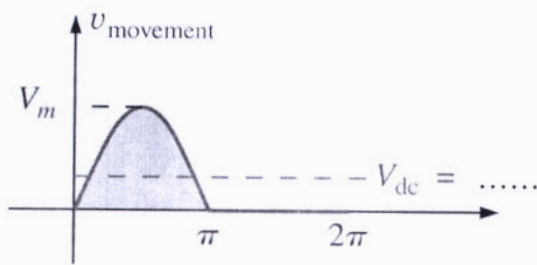
- Answer the questions on this test papers.
- Books, documents and lecture notes are not allowed.
- You must be in the room for one hour after the exam is started and, while taking the exam, you cannot go out except in an emergency case.
- Before leaving, make sure you do not bring this test outside.
- Do not use any electronic communication device.
- Calculators can be used in this test.

Now begin the test.

Cheating in the exam is considered an extremely serious offence which will result in expulsion from the University.

Q1) Find the V_{DC} (V_{Ave}) of a Half-wave rectified waveform by using **Integration**.

(5 marks)



Integration Formulas:

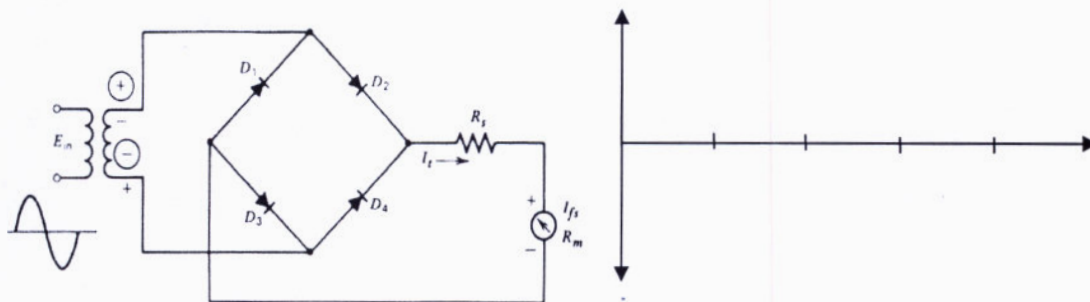
$$\int 1 dx = x + c$$

$$\int \sin x dx = -\cos x + c$$

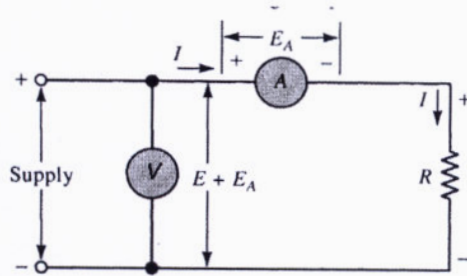
$$\int \cos x dx = \sin x + c$$

Q2) From the figure, **compute** the value of the multiplier resistor (R_s) for a 20-Vrms ac range on the voltmeter and **sketch** the voltage across the meter movement, where a PMMC instrument with FSD (I_{fs}) = 100 μA and meter coil resistance (R_m) is 600 Ω .

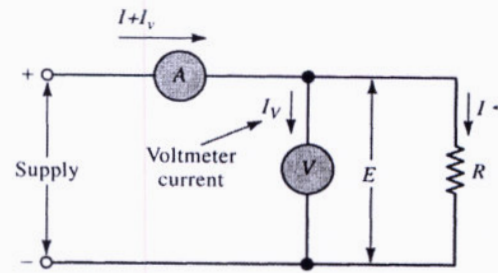
(5 marks)



Q3) From the circuit diagrams, identify each of the voltmeter and ammeter methods and explain their operation of both methods of resistance measurement. (5 marks)



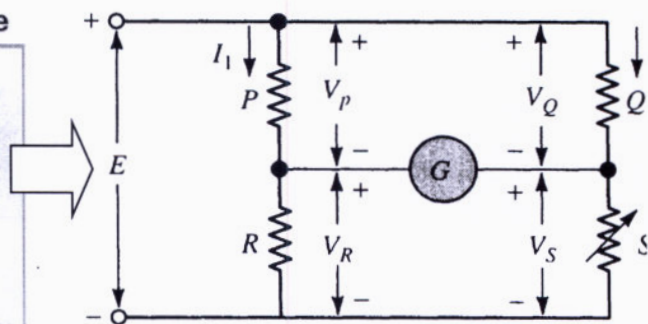
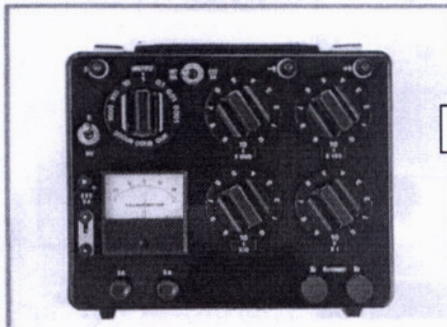
a).....



b).....

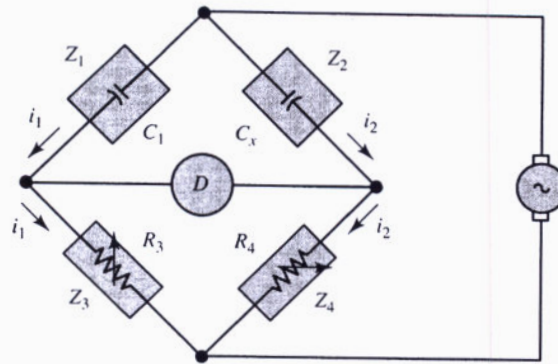
Q4) From the circuit diagram of a wheatstone bridge, explain its operation, and derive its balance equation for R. (5 marks)

Portable Wheatstone Bridge



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Q7) A simple capacitance bridge, as in Figure, uses a standard capacitor (C_1) = $0.01 \mu\text{F}$, and R_3/R_4 can be set to ratio 25:1. Calculate the unknown capacitance C_x . (4 marks)



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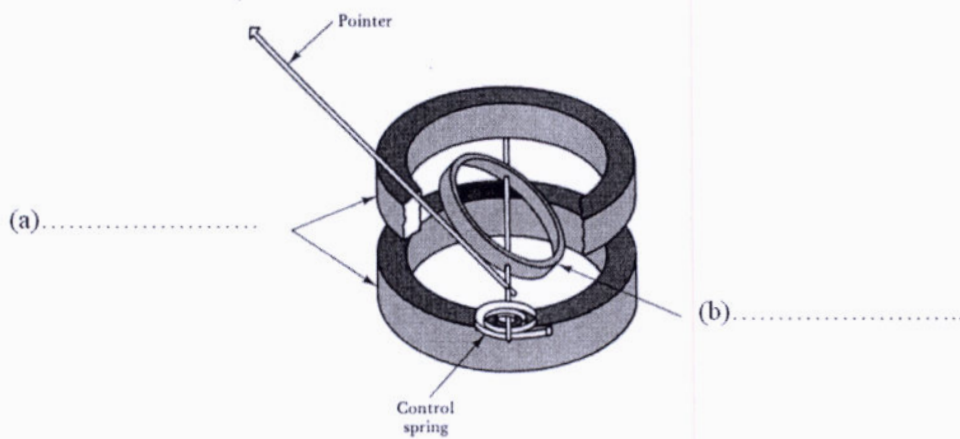
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Q8) Identify each part of the electrodynamic instrument and give the major advantage compare with PMMC. (3 marks)



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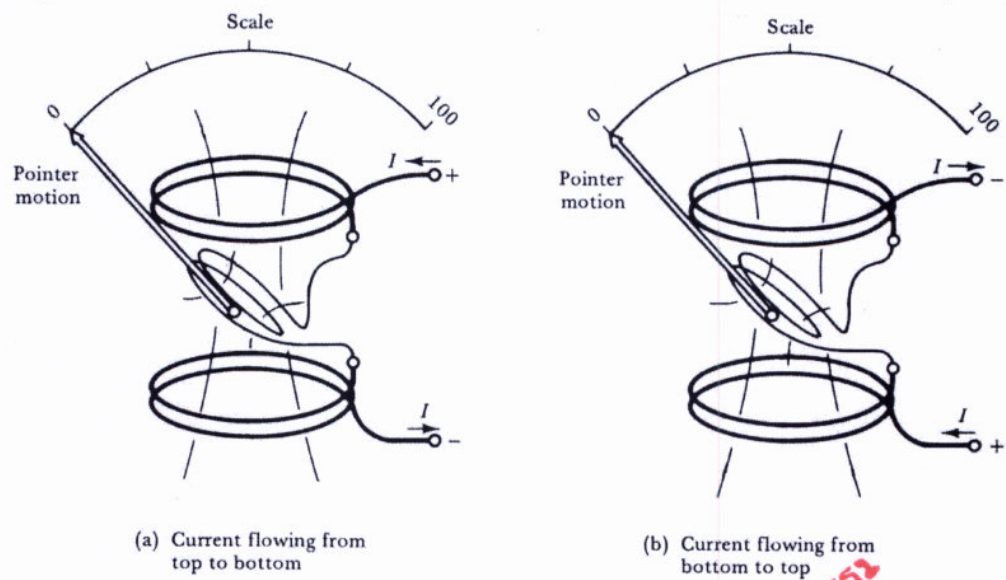
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Q9) Describe the basic AC operation of the electrodynamic instrument as shown in the figure. (4 marks)



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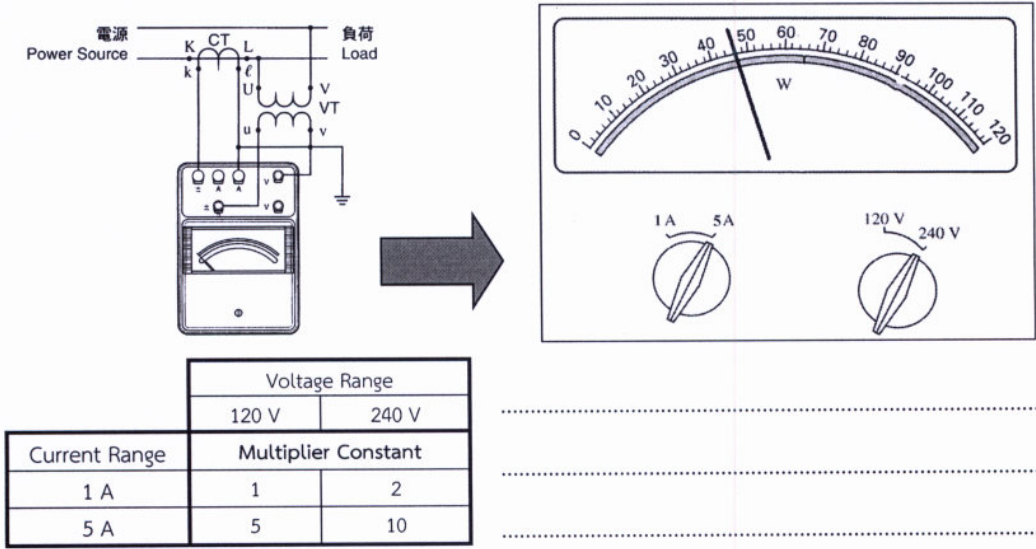
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Q10) For the wattmeter, CT, and VT in Figure, determine the measured power (watt) where: VT ratio = 22000/240 V, CT ratio = 2000/5 A, respectively. (3 marks)



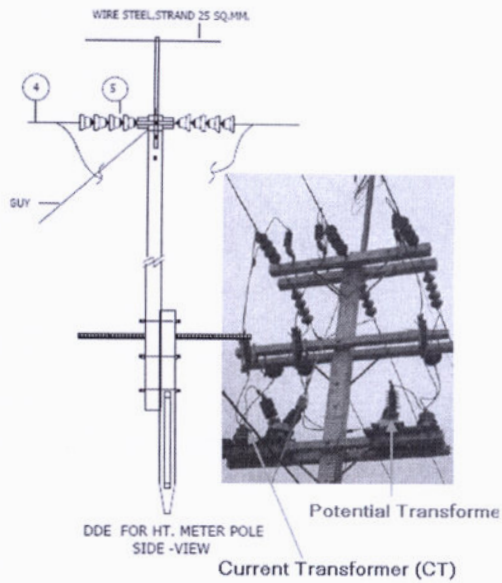
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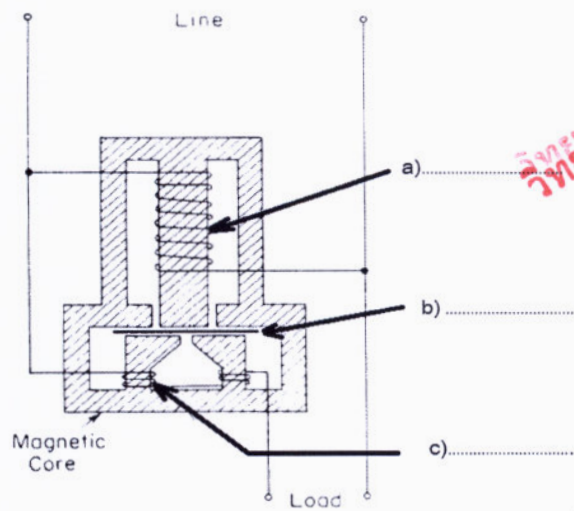
Q11) List main functions of CT and PT.

(3 marks)



Q12) Identify each part of the single phase **watt-hour-meter** and describe its basic concept. (4 marks)

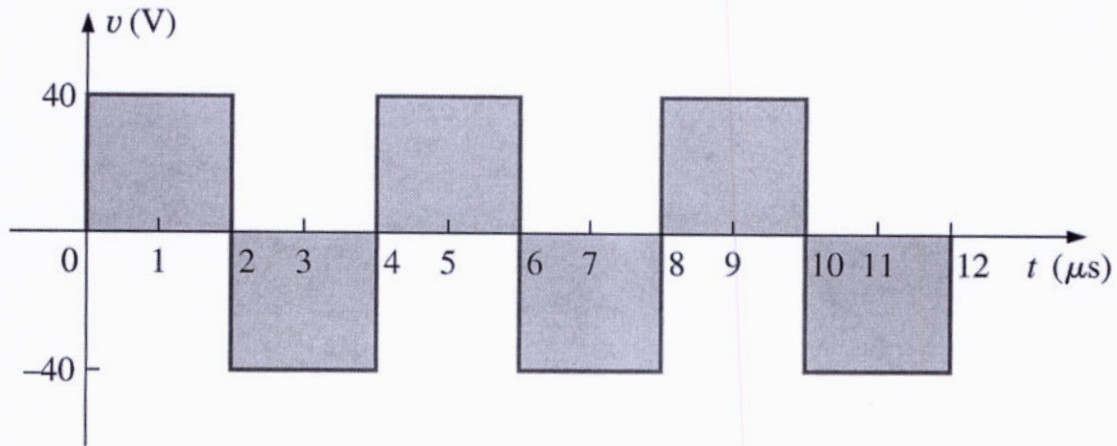
(4 marks)



purchase watt-hour-meter and describe

Q15) Determine the following values of the signal waveform

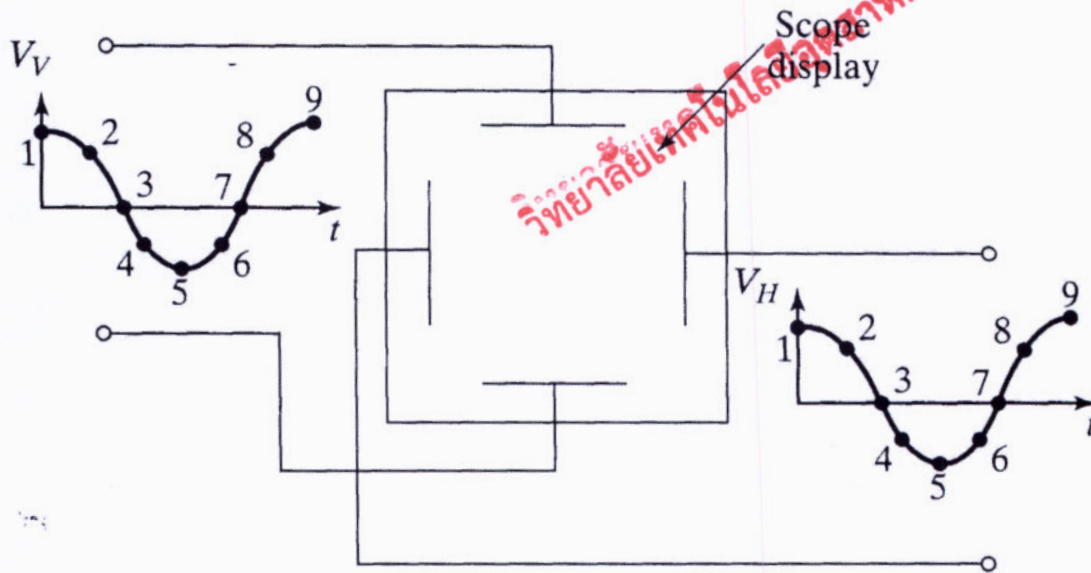
(3 marks)



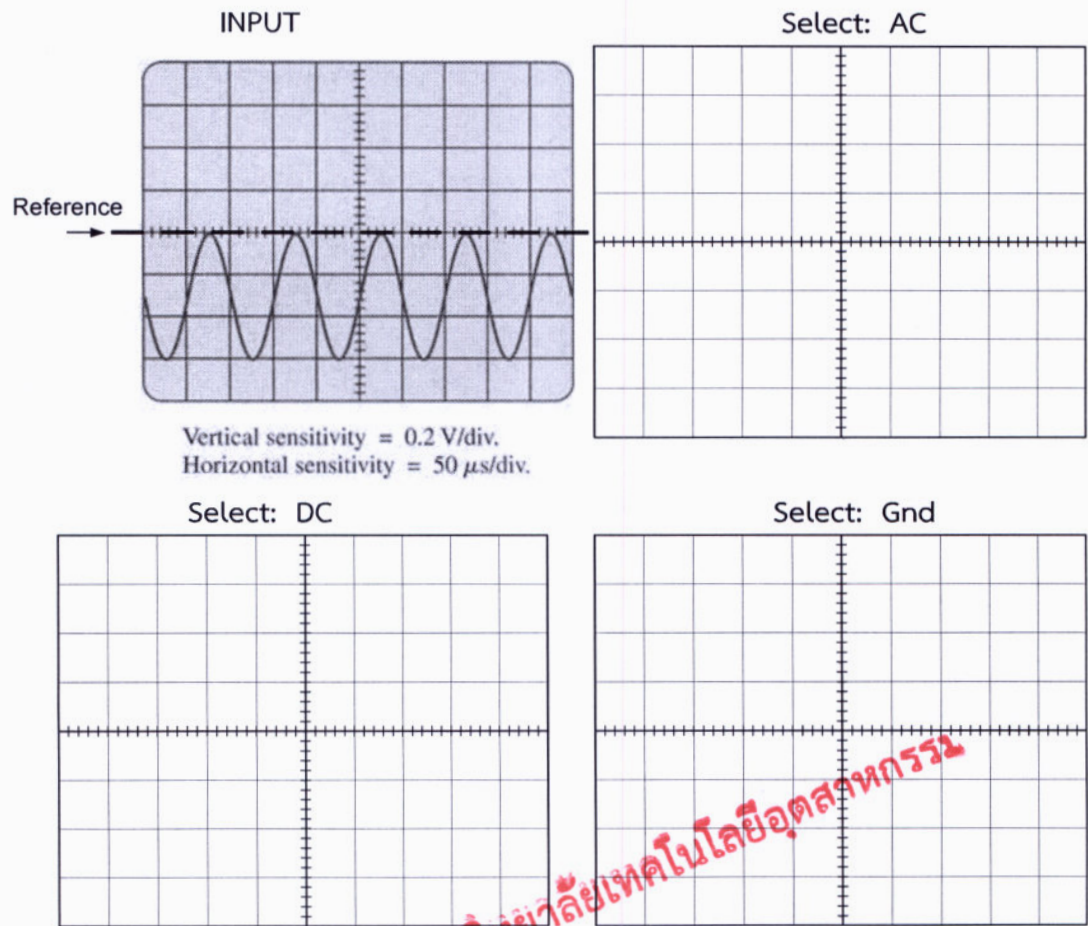
a) Period of the Waveform..... b) Frequency of the Waveform.....

c) How many cycles?

Q16) Draw the scope display that Lissajous patterns are generated with sine waves to both the horizontal and vertical plates. (3 marks)



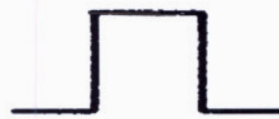
Q17) From the input signal of the oscilloscope, draw the scope display when the setting of input coupling selector to AC, DC and Gnd. (3 marks)



Q18) Describe two figures of probe compensation. (2 marks)



a)



b)

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Q19) Show how to calibrate DC Instrument such as Ammeter using the standard instruments. (3 marks)

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Assoc. Prof. Dr. Pichet Sriyanyong

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